KAIZEN MATHS

Lesson starter

Modulus and Absolute Value

Equations

Name _____

S1

Solve: $|x| = 9$

S2

Solve: $|3x| + 3 = 15$

S3

Solve: $|4x - 7| = 16$

S4

Solve: $|4x - 3| + 7 = 20$

Show your working

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ALG-C00B-L2 · S

Show your working

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Exit ticket

Modulus and Absolute Value

Equations

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E1

A part must be 100mm ± 5mm. Find the acceptable lengths L.

E2

Solve: $|x| = 12$

E3

Solve: $|3x| + 8 = 26$

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ALG-C00B-L2 · E

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Teacher answers & working

Modulus and Absolute Value · Equations

ALG-C00B-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Solve: |x| = 9

Answer:

$x = 9$ or $x = -9$

Working:

Recognize the equation form

|x| = 9 means x is 9 units from zero

Split into two cases

$x = 9$ or $x = -9$

$x = 9$ or $x = -9$

S2

Solve: |3x| + 3 = 15

Answer:

$x = 4$ or $x = -4$

Working:

Isolate the absolute value

$|3x| = 15 - 3 = 12$

Set up two equations

$3x = 12$ or $3x = -12$

Solve each case

Positive case

3x = 12

x = 4

Negative case

3x = -12

x = -4

$x = 4$ or $x = -4$

Teacher answers & working

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S3

Solve: $|4x - 7| = 16$

Answer:

$x = 5.75$ or $x = -2.25$

Working:

Set up two equations

$4x - 7 = 16$ or $4x - 7 = -16$

Solve each case

Positive case

$4x - 7 = 16$

$x = 5.75$

Negative case

$4x - 7 = -16$

$x = -2.25$

$x = 5.75$ or $x = -2.25$

S4

Solve: $|4x - 3| + 7 = 20$

Answer:

$x = 4$ or $x = -2.5$

Working:

Isolate the absolute value

$|4x - 3| = 20 - 7 = 13$

Set up two equations

$4x - 3 = 13$ or $4x - 3 = -13$

Solve each case

Positive case

$4x - 3 = 13$

$x = 4$

Negative case

$4x - 3 = -13$

$x = -2.5$

$x = 4$ or $x = -2.5$

Teacher answers & working

Modulus and Absolute Value · Equations

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E1

A part must be 100mm ± 5mm. Find the acceptable lengths L.

Answer:

$95 \leq L \leq 105$

Working:

Interpret the tolerance

The length may be up to 5mm away from 100mm

Write the absolute value inequality

$|L - 100| \leq 5$

Convert to a range

$100 - 5 \leq L \leq 100 + 5$

Calculate the endpoints

$95 \leq L \leq 105$

E2

Solve: $|x| = 12$

Answer:

$x = 12$ or $x = -12$

Working:

Recognize the equation form

$|x| = 12$ means x is 12 units from zero

Split into two cases

$x = 12$ or $x = -12$

$x = 12$ or $x = -12$

Teacher answers & working

Modulus and Absolute Value · Equations

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E3

Solve: $|3x| + 8 = 26$

Answer:

$x = 6$ or $x = -6$

Working:

Isolate the absolute value

$|3x| = 26 - 8 = 18$

Set up two equations

$3x = 18$ or $3x = -18$

Solve each case

Positive case

$3x = 18$

$x = 6$

Negative case

$3x = -18$

$x = -6$

$x = 6$ or $x = -6$